Experiment 4, Part B: Testing Binder Effects on Ciliary SMO and Hedgehog Signalling

Purpose

Test whether purified MEGF8- and MOSMO-targeting binders alter ciliary SMO and GLI-dependent Hedgehog signalling. A response consistent with interference with MMM regulation would be higher ciliary SMO and higher GLI reporter activity than with a nonbinding control. The binders were designed against human MEGF8 and MOSMO, so activity in NIH/3T3 cells depends on recognition of the corresponding mouse proteins and accessibility of the targeted extracellular epitopes.

Experimental design

- Purified binders are applied to ciliated NIH/3T3 cells, each against a matched nonbinding purified protein and a buffer-only control, at several concentrations so that dose dependence is visible. SMO is read by immunofluorescence against a primary-cilium marker, and ciliary SMO fluorescence is quantified across many cells and independent experiments. Viability and the proportion of ciliated cells are monitored.
- Promising candidates go into Shh-LIGHT2 cells, an NIH/3T3-derived line carrying a GLI-responsive firefly luciferase reporter and a constitutive Renilla control, under the same submaximal SHH stimulation. Firefly normalised to Renilla compares signalling across conditions, and any change is checked for consistency with the ciliary SMO measurement. Gli1 by RT-qPCR gives a second readout where feasible.
- Responses are compared against the nonbinding control and checked for dose dependence. A target-dependent control, or a direct test of whether MEGF8 and MOSMO still associate, is what would show that a response comes from disrupting the complex rather than from something else the purified protein does.

Conditions

Condition	Treatment
Binder	Purified candidate binder at each concentration in the series
Nonbinding control	Matched nonbinding purified protein at the same concentrations. This protein will be identified from the mammalian two-hybrid screen as one that has the least-off target signal but also shows no binding as compared to the control
Buffer only	PBS, the buffer used for the purified proteins

Binder concentration series: a half-log series from 1 nM to 1 μM (1, 3, 10, 30, 100, 300 and 1,000 nM). Keep the PBS volume the same in every well and no more than 10% of the medium volume.

Materials and equipment

- Purified binders and nonbinding control protein in PBS, at standardised concentrations
- NIH/3T3 cells and Shh-LIGHT2 cells
- 24-well plates with glass coverslips; white 96-well plates for luminescence

- Low-serum medium; complete medium for both cell lines (NIH/3T3 complete medium as defined in Experiment 3; Shh-LIGHT2 complete medium as recommended by the supplier); selection antibiotics for Shh-LIGHT2 maintenance
- SHH stimulus: recombinant SHH-N or SHH-conditioned medium
- Primary antibodies: rabbit polyclonal anti-SMO (1:1000) and anti-ARL13B (1:1000); Alexa Fluor secondary antibodies
- Mounting medium with DAPI; fluorescence microscope; Fiji
- Dual-Luciferase Reporter Assay System (Promega) and a luminometer

Procedure: SHH calibration

1. Before the experiment, titrate SHH on Shh-LIGHT2 cells and choose the dose that gives a low, sub-saturating response, less than about 10% of the maximum. Use this dose in every imaging and reporter well.

Procedure: ciliary SMO in NIH/3T3 cells

1. Plate 100,000 NIH/3T3 cells per well on glass coverslips in a 24-well plate in complete medium and grow them to confluence.
2. When the cells are confluent, replace the medium with low-serum medium (complete medium with 0.5% FBS) containing the binders, nonbinding control or buffer-only control at the chosen concentrations, together with the same submaximal SHH stimulus used in the GLI reporter assay. Incubate for 24 hours; primary cilia form during this period.
3. Fix in chilled 4% paraformaldehyde in PBS for 10 minutes on ice, rinse three times in PBS, and block for 1 hour in 1% horse serum and 0.1% Triton X-100 in PBS.
4. Incubate with primary antibodies against SMO (1:1000) and ARL13B (1:1000) overnight at 4 °C, then with secondary antibodies for 1 hour at room temperature. Mount with an antifade medium containing DAPI.
5. Image with a 60× oil objective, acquiring z-stacks about 4 µm thick with the same acquisition settings for every condition within the experiment.
6. In Fiji, make maximum intensity projections, build ciliary masks from the ARL13B signal, apply them to the SMO channel, and subtract the mean intensity of a cell-free background region.
7. Analyse about 100 cilia per condition, in three independent experiments.
8. Record viability from the DAPI channel, as nuclei number and nuclear morphology per field, and record the proportion of ciliated cells for every condition.

Procedure: GLI reporter in Shh-LIGHT2 cells

1. Maintain Shh-LIGHT2 cells in DMEM with 10% calf serum and the selection antibiotics, as recommended by the supplier.
2. Seed cells in white 96-well plates at 10,000 cells per well and grow them to confluence.
3. Switch to low-serum medium (DMEM with 0.5% calf serum) containing a submaximal SHH stimulus and the binders, nonbinding control or buffer at the chosen concentrations.
4. Incubate for 24 hours, as for the imaging.
5. Lyse the cells and measure firefly and Renilla luciferase with the Dual-Luciferase assay.
6. Calculate firefly divided by Renilla for each well and compare across conditions. Renilla signal also serves as an internal check on cell number and viability. Check whether any change is consistent with the ciliary SMO result.
7. Repeat in three independent experiments.

Controls

- Matched nonbinding purified protein at the same concentrations.
- Buffer-only control.
- Dose dependence across the concentration series.
- Viability and proportion of ciliated cells in every condition.
- The same submaximal SHH stimulus in all imaging and GLI reporter wells.

Expected readout and analysis

- A candidate that interferes with MMM regulation should raise ciliary SMO fluorescence relative to the nonbinding control, and under a matched submaximal SHH stimulation should raise normalised GLI reporter activity as well. A dose-dependent response in both would support functional inhibition of MMM-mediated regulation. It would not, on its own, establish direct disruption of the MEGF8 and MOSMO interaction, which would require a separate test of whether the two proteins still associate.
- A binder that raises ciliary SMO without raising GLI activity would indicate that more SMO in the cilium does not automatically produce more signal.
- A candidate that does nothing may have failed to reach or recognise the mouse target, or may be binding without disrupting anything. The functional assay alone cannot separate those, and a flat result would not establish that a binder is nonfunctional against its intended human target.

References

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